

Viewpoint

Real time public display dashboards: Reframing road traffic accidents in India

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India's escalating number of road traffic accidents, demands systemic transformation beyond traditional enforcement measures. This viewpoint proposes the establishment of a public, real-time Trauma Dashboard to transform fragmented, delayed reporting into an open, evidence-based system of accountability. The dashboard would integrate cross-verified datasets from police, hospitals, and insurance companies through interoperable digital platforms such as Open Data Kit (OTK), and Talend Open Studio, or similar platform hosted on the Government of India's MeghRaj Cloud. Automated cross-verified data flows would provide dynamic insights into crash patterns, high-risk zones, and temporal trends. Drawing on successful precedents such as the Central Pollution Control Board (CPCB) Air Quality Index and trauma registries - UK Trauma Audit and Research Network (TARN), US National Trauma Data Bank (NTDB) and India's National Organ and Tissue Transplant Organization (NOTTO), this project proposes to facilitate continuous monitoring, policy responsiveness, and public awareness.

Keywords CPCB AQI dashboard; Dashboards; Mortality and morbidity of trauma; Public-private-partnership; Road traffic accidents; Trauma

India loses over 150,000 lives annually to road traffic accidents.¹ These are reported as isolated events rather than indications of systemic failure in urban planning, enforcement, and response. Without aggregated visibility, road trauma counts among India's most under-recognised public health crises. The annual total of those who die on Indian roads or 1.19 million worldwide as estimated by the World Health Organization is a major public health issue.²

This viewpoint proposes the creation of a real-time Trauma Dashboard to regularly record and display road accident data. A Dashboard has the potential to highlight preventable deaths, guide enforcement, and foster accountability at the local, state, and national levels.

Advocacy to implement road safety laws globally

International practice has shown that regular reporting, citizen participation, and transparent data can drive

sustainable road safety reforms. The EU Road Safety Exchange (2020–2025) connects countries to share best practices towards the Vision Zero aim to eliminate fatalities by 2050.³ In Sweden, Vision Zero policy combines data-driven interventions and public display Dashboards for measuring progress.⁴ In Australia, coalitions among health agencies and media exist to enable contextual accident reporting rather than singular incidents.⁵

In India, civil society initiatives - the SaveLIFE Foundation and policy actions such as the Good Samaritan Law demonstrate the potential of evidence-based advocacy to trigger policy change.⁶ Much of India's accident data is not standardised, and in some instances available only upon request, which limits awareness and compliance. A real-time Trauma Dashboard, built through a public-private partnership (PPP) model, may combine these streams of data and give an open window into avoidable deaths, translating advocacy into governable action.

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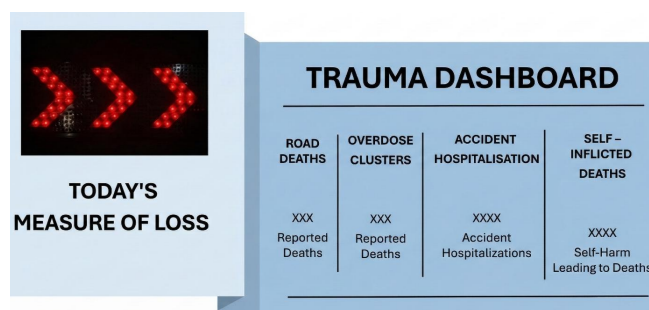


Figure. Outline of proposed Dashboard.

Reliability of data input into a proposed Trauma Dashboard

The reliability of a Trauma Dashboard will be dependent on the quality and reliability of its input data (**Figure** shows proposed outline). Trauma data is obtained from a variety of sources - emergency departments, police blotters, and insurance claims, each with differing formats, reporting standards, and timeliness. Without convergence, gaps and inconsistencies could dilute the system's credibility. To avoid this, standardised reporting protocols, interfaces with electronic health records, and independent validation systems must be instituted.

India's own domestic experience with air-quality dashboards is instructive. The Central Pollution Control Board (CPCB) employs calibrated sensors, standardised air quality index (AQI) methodology, and automated real-time feeds to minimise the scope for human judgment and allow cities to be comparable to each other (CPCB AQI Dashboard).⁷

India collects a plethora of data on road accidents from multiple agencies such as the Ministry of Road Transport and Highways (MoRTH), National Crime Records Bureau (NCRB), State traffic police, and insurance companies. However, the information are not centralised and real-time; data are scattered among departments and are generally released after cumbersome application process or in annual reports.

This lack of accessible, aggregated data seriously impedes public awareness, timely analysis, and evidence-based policymaking. Developing a real-time, publicly visible Dashboard would mobilise this dormant data as an active instrument of accountability and prevention.

Global precedents in trauma data systems

Other nations provide examples of how real-time Trauma Dashboards can revolutionise tracking and

policy response. UK Trauma Audit and Research Network (TARN) provides a national trauma registry and performance Dashboard that enables hospital-level benchmarking of outcomes, influencing care quality and redesign of the trauma system.⁸ Similarly, in the US, National Trauma Data Bank (NTDB) and National Trauma Quality Improvement Program (TQIP) serve as real-time reference systems for the occurrence of trauma and improvement in care.⁹ India's National Organ and Tissue Transplant Organization (NOTTO) dashboard shows how public, real-time, transparent registries can engage different stakeholders with digital governance.¹⁰

A similar model can be utilised for tracking road trauma, linking ambulances, hospitals, and police departments into one network for real-time reporting. Implications from such systems can be used in developing an Indian Road Trauma Dashboard with a focus on interoperability, measurement-based data collection, and real-time feedback loops.

Public - Private Partnership (PPP) model

The proposed Dashboard can adopt a PPP model in both innovation and sustainability.

Public stakeholders: MoRTH, National Informatics Centre (NIC), and State Health Departments could handle data governance and standards.

Private partners: Select technology firms, academic analytics labs, and NGOs can supply data integration, visualisation, and AI analytics.

Outcome: Such a partnership would ensure credibility, cost-sharing, and rapid rollout, as in CPCB's air quality monitoring network or NOTTO's organ donation registry system.

The multiple nodes on the Dashboard could integrate data from traffic police accident reports, hospital emergency and ambulance logs, and insurance company databases through standardised APIs (Application Programming Interface). It would be presented by districts as well as the State, updated hourly, and published publicly on web, billboards, and mobile interfaces. The architecture could enable real-time public awareness. The proposed data collection and flow for a National Trauma Dashboard is shown in **Table**.

Technology platform and software framework

To implement the suggested Trauma Dashboard, an open-source, modular software stack can be utilised. The

Table. Proposed data collection and flow for a National Trauma Dashboard (India)

Level	Primary data source	Key data elements collected	Data validation and transmission	Central integration / End-user access
Local (District / City)	Traffic police, emergency ambulance services, hospital emergency departments	Time and location of crash, GPS coordinates, number of persons injured/deceased, vehicle type, cause (speeding, alcohol, road condition), emergency response time	Automated digital entry <i>via</i> mobile/tablet; GPS tagging and timestamp; daily upload to State Dashboard through secure API	—
State / Regional	State health departments, transport departments, police traffic wing	Aggregation of district data; injury severity classification; hospital capacity/utilisation	Data cleaning, de-duplication, and verification by nodal officer; automated sync to National Dashboard	—
National (Central repository)	MoRTH iRAD, ICMR-NIC collaboration	Consolidated dataset covering all states; analytics on trends, demographics, and hotspots	AI-based anomaly detection; cross-validation with NCRB and insurance records	Public Trauma Dashboard (web and mobile); access for policymakers, media, researchers
Public and policy feedback loop	Citizens, civil society, researchers, media	Real-time visualisation, policy alerts, open datasets	Feedback channels for corrections and suggestions	Drives awareness, accountability, and evidence-based enforcement
MoRTH, Ministry of Road Transport and Highways; iRAD, Integrated Road Accident Database; ICMR-NIC, Indian Council of Medical Research - National Informatics Centre; APIs, application programming interface; GPS, global positioning system				

aim is to have interoperability, minimal implementation cost, and rapid scalability across metropolitan cities and beyond as shown in **Supplementary Table I**.

Automated checks will be done through the (NIC-hosted MeghRaj Cloud infrastructure with Ministry of Road Transport and Highways - Integrated Road Accident Database (MoRTH-iRAD) and Indian Council of Medical Research - National Informatics Centre (ICMR-NIC) collaboration datasets. Data packets routed from State Dashboards could go through multi-layer checks like de-duplication, timestamp and geo-tag cross-verifications, and schema integrity with Talend Data Quality modules. Artificial-intelligence-based algorithms will detect outliers or inconsistencies by comparison with National Crime Records Bureau (NCRB) crash reports, insurance claim database information, and hospital admission information through the National Health Stack application programming interface (API). Cryptographic checksums will provide integrity of transmission, and nodal officers could perform regular manual verification to reconcile state-level and national aggregates before releasing these publicly on the Trauma Dashboard.

Public interface and visibility

The proposed Trauma Dashboard must be designed as a multi-platform public information system, highly visible over digital, institutional, and physical

interfaces in order to achieve maximum awareness and policy sensitivity. **Supplementary Table II** gives the proposed methods of public display of trauma data.

We propose that the Trauma Dashboard should be placed in the public domain through a national and State-level web portal, supported by mobile applications and embeddable media devices. Additional citizen involvement can also have live Dashboards displayed on electronic billboards at key junctions and transport hubs. This multi-channel launch will ensure that road safety information is not restricted to policymakers, but included as part of the public's daily awareness, similar to air-quality or weather dashboards.

Overall, a Trauma Dashboard is not only a helpful technical tool as an agent of change; it converts scattered accident reports into system analysis that reveals patterns of risk and disparity. In the process, it makes them transparent to enable civil society and media to put pressure for enhanced enforcement of road safety laws and mobilise public opinion for reforms. We recommend that the establishment of a Trauma Dashboard must be followed by its governance - transparency of data collection, and display. Furthermore, the intent of such a Dashboard is prevention; aligning trends in the data with targeted safety measures such as improved road design, enforcement of helmet and seatbelt use and anti-drunk driving measures.

We believe that the display of road deaths, suicides, and accidents on the same page as weather reports and stock market indices to sensitise citizens that safety is measurable. Research will have to be carried out to assess the impact of the proposed Trauma Dashboard, initially in a geographically defined urban city, followed by a State and eventually India-wide. AI enabled technological advances may enhance our ability to collect, validate, and disseminate real time data.

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